Remarks

We note that

- not all solutions of the derived polynomial equation give a solution for the given equation;
- there may be infinitely many solutions for non-polynomial equations though they look like polynomial equations;
- there may be no solution for such equations.
- the Fundamental Theorem of Algebra is proved only for polynomials; for non-polynomial expressions, we cannot talk about degree and hence we should not have any confusion on the Fundamental Theorem of Algebra having non-polynomial equations in mind.

EXERCISE 3.5

1. Solve the following equations

   (i) $\sin^2 x - 5\sin x + 4 = 0$ (ii) $12x^3 + 8x = 29x^2 - 4$

2. Examine for the rational roots of

   (i) $2x^3 - x^2 - 1 = 0$ (ii) $x^8 - 3x + 1 = 0$.

3. Solve : $8x^{\frac{3}{2n}} - 8x^{\frac{-3}{2n}} = 63$

4. Solve : $2\sqrt{\frac{x}{a}} + 3\sqrt{\frac{a}{x}} = \frac{b}{a} + \frac{6a}{b}$.

5. Solve the equations

   (i) $6x^4 - 35x^3 + 62x^2 - 35x + 6 = 0$ (ii) $x^4 + 3x^3 - 3x - 1 = 0$

6. Find all real numbers satisfying $4^x - 3(2^{x+2}) + 2^5 = 0$.

7. Solve the equation $6x^4 - 5x^3 - 38x^2 - 5x + 6 = 0$ if it is known that $\frac{1}{3}$ is a solution.

3.9 Descartes Rule

In this section we discuss some bounds for the number of positive roots, number of negative roots and number of nonreal complex roots for a polynomial over $\mathbb{R}$. These bounds can be computed using a powerful tool called "Descartes Rule".

3.9.1 Statement of Descartes Rule

To discuss the rule we first introduce the concept of **change of sign in the coefficients** of a polynomial.

Consider the polynomial.

$$2x^7 - 3x^6 - 4x^5 + 5x^4 + 6x^3 - 7x + 8$$

For this polynomial, let us denote the sign of the coefficients using the symbols '+' and '−' as

$$+, -, -, +, +, -, +$$

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